Applied Economic Analysis (Econ 300)

Fall 2025

Time and venue: MW 2PM-3.30PM, SH 328

Instructor: Dr. Aakanksha Melkani (amelkani@siena.edu; (518)-783-2929)

Office: Kiernan Hall 102D

Office hours: MW 11.30AM-12.30PM; T 10AM-1PM

Course Description

The course is aimed at fostering familiarity and understanding of key statistical and mathematical concepts that are fundamental for intermediate and advanced training in economics. Analytical tools such as MS Excel will be used throughout the semester to complement conceptual understanding with practical application and to provide hands-on training in these analytical tools. This course has no prerequisites.

Assessable Learning Objectives

1. Demonstrate understanding of relevant statistical and mathematical techniques used in economic modelling
2. Capability to recognize technical statistical and math concepts and apply them to economic ideas
3. Confidence in using MS Excel for basic data wrangling and analysis

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Course Materials

We will use several free online resources for this course. Some of these are listed below. More might be shared as the course progresses. Please refer to class notes on Canvas for topic specific resources.

Excel: https://www.coursera.org/projects/introduction-microsoft-excel

Statistics: https://online.stat.psu.edu/stat200/lesson/1/1.2-

https://onlinestatbook.com/2/index.html

https://openstax.org/details/books/introductory-statistics-2e

Math: https://calculusmadeeasy.org/

Assessment and Grading

Type	Detail	Weight
Pop-quizzes	Automatically graded on Canvas. Unlimited attempts allowed (till the last day of class).	20%
Class participation	Several small projects to be finished within the class. Group work encouraged though not necessary. Submission must be individual. If you miss a class, you miss the grade. Includes test on last day of class	20%
Mini projects	There will be several mini projects throughout the class. Most will require MS Excel. Each mini project will be graded in two parts: 1. Homework: The project will be assigned to you as homework, and you will try to figure out the solution to the best of your capability. Half the grade will be assigned to this component. 2. Class work: All projects will be solved in class after you have attempted them as homework. Half the grade will be based on your timely completion of the project after it has been discussed in class.	60%

Grades: Final grades will be determined based on the following break-up.

		B+	85 to < 90	C+	72 to < 75	D+	62 to < 68	F	<50
A	>= 94	B	80 to < 85	C	70 to < 72	D	58 to < 62		
A-	90 to < 94	B-	75 to < 80	C-	68 to < 70	D-	50 to < 58		

Course outline

Topic	Dates	Details
PART I: Intro to MS Excel and descriptive stats for Economics	Sep 3	Introduction, Syllabus, MS Excel setup
	Sep 8 and 10	Descriptive stats - Central tendency, dispersion, and relation.
	Sep 15 and 17	Data distributions (qualitative and quantitative variables). Deciles and percentiles, 5-number summaries.
PART II: Inferential Stats	Sep 22 and 24	Key concepts: Population and Sample; Probability; Sampling distributions. Intuitive intro to inferential stats
	Sep 29 and Oct 1	Point and interval estimates; Intuitive introduction to hypothesis testing
	Oct 6 and 8	Hypothesis testing and statistical significance
	Oct 13 and 15	In-class practical on summarizing statistical results and hypothesis testing. Replication assignment for the fall break
	Oct 20 and 22	No class – work on replication assignment
PART III: Math for Economics	Oct 27 and 29	Solve replication assignment in class. Introduction to functions and their graphical representation
	Nov 3 and 5	Classical linear functions in Economics. Solving a system of equations
	Nov 10 and 12	Classical non-linear functions in Economics
	Nov 17 and 19	Derivatives: Intuitive and technical; Concavity and convexity
	Nov 24	Derivatives – In-class exercise
	Dec 1 and 3	Derivatives – Optimization and Marginal analysis.
Last class	Dec 8	In class test

Communication

Apart from the classroom instruction, I will extensively use your Siena email id and Canvas to communicate with you. **PLEASE check your email and Canvas regularly for new communication and announcements.** You are encouraged to communicate and seek guidance with the instructor, especially during the office hours. You can also communicate through email and setup appointments if the office hours do not work for you. I will try to reply to all emails within 24-hours during working hours (Monday to Friday, 8AM to 5PM). My responses during other times may be slow so please plan your work ahead and do not keep urgent questions for the weekend.

Class Policy around the use of Artificial Intelligence (particularly Chat-GPT)

You may use all online resources accessible to you to help you understand the class material. **However, all assignment submissions should be your own individual effort.** Copying and pasting text from online resources, including Chat-GPT, can be considered plagiarism, which is a case of academic dishonesty. Please read https://www.siena.edu/offices/academics/academic-policies-and-procedures/academic-integrity-policy/ to familiarize yourself with the College academic integrity regulations

School of Business Mission Statement

We offer a business education integrated with the liberal arts and rooted in the Franciscan values of inclusivity and service. Our AACSB-accredited undergraduate and graduate programs develop versatile and socially responsible citizens with the skills to compassionately lead, serve, and succeed.

School of Business Learning Outcomes

The School of Business provides intellectually challenging educational programs in Accounting, Economics, Finance, Management, and Marketing that empower our students to address the challenges of an ever-changing world and a diverse society. Each School of Business graduate will attain the following outcomes:

1. Decision Making - Students will make and implement sound decisions.

2. Communication - Students will demonstrate effective communication skills across platforms and audiences.
3. Teamwork and Leadership - Students will demonstrate effective leadership and team member skills.
4. Social Responsibility - Students will recommend actions that consider the impact on all stakeholders, the external environment, and the natural world.
5. Personal Development - Students will have the foundation to build personal and career success.
6. Business Specialization - Students will address organizational challenges and opportunities by applying the in-depth perspective of business specialization

Economics Department Mission

The mission of the Economics Department at Siena College is to educate undergraduate students in a learning environment that emphasizes both mainstream and heterodox approaches to economics, as well as principles of economic justice related to Siena's Franciscan tradition. The Economics Department offers a curriculum that is taught critically and comparatively to promote inquiry and intellectual growth for students, department faculty, and the campus community. The B.A. offers students the opportunity to focus on the relationship between economics and society, while the B.S. requires training in functional areas of business.

Economics Department Learning Goals

-*Concept mastery*: Show mastery in central concepts of mainstream and heterodox approaches to economics.
- *Inquiry:* Provide critical and comparative inquiry through engagement with texts, models, and data. (Includes critical thinking, problem solving, and data analysis.)
- *Policy analysis:* Analyze policy with respect to mainstream and heterodox approaches to economics, and principles of economic justice related to Siena's Franciscan tradition.
- *Communication:* Demonstrate intellectual growth by communicating inquiry-based findings